1. Talk 1 (15.10.2026, Kasra). Why do formal group laws arise?

Two main references are: Lecture 1 [1], [2] for details about complex characteristic classes, the classifying space $BU(n)$ and complex K-theory.

- Define Chern characteristic classes, discuss operations on the set of vector bundles.
- State the existence of the classifying space $BU(n)$ and show a sketch of the computation of its cohomology
- Define complex-oriented and even cohomology theories, explain their importance
- Work out two main examples: complex K-theory and integral cohomology, explain how formal group laws arise in these two examples
- Define a formal group law

2. Talk 2 (22.10.2026, Maqbool). Equivalences of Formal Groups

- Briefly outline the picture from differential geometry: the beginning of Lecture 2 [1] and I.'s notes or any other source.
- Introduce étale sheaves as geometric setting, follow Lecture 2 [1], skip the part about the category of elements: [Definition 2.1], [Remark 2.2], some examples, [Definition 2.16] and below.
- Define formal groups and show how formal group laws produce formal groups, follow Lecture 3 [1]: [Definition 3.1], [Proposition 3.5], only if time permits, [Definition 3.7], [Lemma 3.9], [Definition 3.14]
- Work out the example of complex-oriented cohomology theory and KU if time permits: [Example 3.17] and [Remark 3.18] [1], respectively. Show how $\mathbb{G}_m$ and $\mathbb{G}_a$ arise as formal groups associated to KU and $H\mathbb{Z}$ respectively.
- Sketch that over $\mathbb{Q}$ all formal group laws are equivalent: [Theorem 5.13] [1] using an intuition from differential geometry and taking some statements on faith (such as [Lemma 5.2] and [Lemma 5.11]).

3. Talk 3 (29.10.2026, Gun). Moduli of Formal Groups

All references are from [1].

- Representability of functors, show moduli problems (for FG and FGL) representable, and what they are represented by.
- Define the functor $\mathcal{F}\mathrm{gl}$ and prove that it is represented by the Lazard ring [Lemma 6.1, Definition 6.2]. State Lazard's theorem without proof [Theorem 6.8] and explain how the proof over $\mathbb{Q}$ follows from [Theorem 5.13]: [Corollary 6.10], and deduce that formal group laws can always be lifted along surjective homomorphisms [Corollary 6.9].
- Define the moduli stack of formal groups $\mathcal{M}_{\mathrm{fg}}$ following [Definition 12.1]. State and prove [Prop. 12.2, Prop. 12.3]. Define sheaves with values in anima [Definition 8.1]

4. Talk 4 (Rescheduled! Wed 04.11.2026, 10:00–12:00, Aila). Complex Orientability & MU is the Universal Complex Oriented Theory

All references are from [1].

- In this talk, we will introduce the cohomology theory MU of complex bordism and show that MU is universal among complex-oriented cohomology theories.
- Briefly recall what an E_∞-ring spectrum is.
- Define MU as an E_∞-ring spectrum. State and prove [Theorem 9.14].
- State Quillen's theorem [Theorem 9.15, Theorem 11.8] + (Remark 9.11) and explain its consequence following [Theorem 11.10] and the paragraphs before and after it if time permits.

5. Talk 5 (12.11.2026, Lasse). Formal Groups Classify Certain Cohomology Theories

- State and prove [Proposition 12.8], together with [Proposition 12.2] and [Proposition 12.3].
- Hence deduce that we have a map

$$\mathrm{Sp} \longrightarrow \mathrm{QCoh}(\mathcal{M}_{\mathrm{fg}})$$

following [Rognes, Definition 11.5.2], [Theorem 11.5.1] as well.

6. Talk 6 (19.11.2026, Li). Frobenius and heights

- Define the Frobenius automorphism [Definition 13.1] [1]
- Define what it means to have height at least n [Definition 13.8] [1] + explain its connection with a "usual" definition (see for example Lecture 12 [3]) + [Example 13.16] [1]
- Explain that not all formal groups have a height [Definition 13.11] and below, [Corollary 13.13]
- Define the elements v_n [Definition 13.17] [1] + [Corollary 13.19] [1]
- Introduce the height filtration on the moduli stack of formal groups [Proposition 14.1] + [Proposition 14.3] [1]

7. Talk 7 (26.11.2026, Nikita). How to go from Formal Groups to Cohomology Theories: Forms of K-Theory

- Show how to construct cohomology theories from formal groups: Construction 19.1 [1]
- Formulate the Landweber Exact Functor Theorem: Theorem 19.5 [1]
- Show how deformation theory is used to construct Landweber exact sequences: Corollary 18.19, Lemma 20.21, Definition 20.22, Theorem 20.23

8. Talk 8 (03.12.2026, William). Intro to Ring Spectra and Cohomology Theories E_∞, E_1

9. Talk 9 (10.12.2026, Gautham). Spectra are Quasicoherent Sheaves on the Moduli Stack of Formal Groups

Supposed to be covered: *(Relation between Sp and QCoh(M_{fg})) Explain how to extend the action in inf-cat. terms, Spec S and Spec Laz iso, mention that by a module constructed above we obtain a sheaf on Spec L, upgrade it to define map Sp $\longrightarrow$ QCoh(M_{fg}), state nilp theorem and what it tells us about the map.*

10. Talk 10 (17.12.2026, Sayan). Spectral Primes: Morava E-theory and Morava K-theory

Supposed to be covered: *Constructing Morava E-theory and showing it is E_∞, Morava K-theory as E_1 ring. Using LEFT, introduce Lurie's obstruction theory theorem (faithful flatness)*

11. Talk 11 (07.01.2027, Thorger). Forms of K-theory: Getting to Know the First Elliptic Cohomology

Supposed to be covered: *Hands on $E(1) = KU_p^\wedge$. Description of $L_{K(1)}S$. Illustrate on this example the properties covered in the previous talks.*

12. Talk 12 (14.01.2027, Priyanka). Chromatic Fracture Square

Supposed to be covered: *The Chromatic Fracture Square. Bousfield localisation, $L_{E(n)}$ is a smashing localisation. Height filtration.*

13. Talk 13 (21.01.2027, Phil). E-theory is a Galois cover of the $K(h)$-Local Sphere

Supposed to be covered: *The Devinatz–Hopkins Theorem*

References

[1] P. Pstrągowski, *Finite Height Chromatic Homotopy Theory*, Lecture notes for Math 252Y, Harvard University, Spring 2021.

[2] J. Rognes, *Chromatic Homotopy Theory*, Algebraic Topology III, MAT4580/MAT9580, University of Oslo, 2023.

[3] J. Lurie, *Chromatic Homotopy Theory*, lecture notes for Math 252, Harvard University.